

ASCII MASTER FLOW – PANEL 1/12: Coastal-system balance

WAVES + TIDES + CURRENTS -> marine energy

RIVERS + CLIFFS + OFFSHORE -> sediment supply

SEA LEVEL + LAND MOTION -> accommodation space

HUMAN WORKS -> redistribute erosion and accretion

MUST REMEMBER: Coastal morphology connects wave generation, refraction, erosion, longshore drift, sediment cells, deposition and sea-level/tectonic change to cliffs, caves-arches-stacks, beaches, spits, bars, lagoons, deltas and India's east-west coast contrast.

ASCII MASTER FLOW – PANEL 2/12: Refraction map

DEEP WATER -> wave fronts travel faster

SHALLOW HEADLAND -> fronts slow and converge

BAY -> fronts diverge and energy disperses

RESULT -> erosion focus / depositional shelter

ASCII MASTER FLOW – PANEL 3/12: Erosional sequence

JOINT / FAULT -> hydraulic action + abrasion

CAVE -> widening and roof attack

ARCH -> roof collapse -> STACK

STACK -> undercutting -> STUMP

ASCII MASTER FLOW – PANEL 4/12: Cliff retreat section

WAVE NOTCH -> basal undercutting

OVERHANG -> instability and collapse

DEBRIS REMOVAL -> renewed attack

RETREAT -> widening wave-cut platform

ASCII MASTER FLOW – PANEL 5/12: Depositional discriminator

SPIT -> one end attached

BAR -> crosses bay or mouth

LAGOON -> water enclosed behind bar

TOMBOLO -> joins island to land

ASCII MASTER FLOW – PANEL 6/12: Relative sea-level fork

LAND RISE / SEA FALL -> emergence

RAISED BEACH + TERRACE -> exposed marine forms

LAND FALL / SEA RISE -> submergence

RIA = drowned river valley / FJORD = glacial trough

ASCII MASTER FLOW – PANEL 7/12: Storm-surge chain

CYCLONE WIND + LOW PRESSURE -> water setup

TRACK + SHELF + BATHYMETRY -> local amplification

HIGH TIDE + WAVES + RIVER FLOW -> compound level

EXPOSURE + WEAK EVACUATION -> disaster risk

ASCII MASTER FLOW – PANEL 8/12: India coast contrast

EAST -> broader plains + large deltas

WEST -> narrower plain + many estuaries

PORT TEST -> depth, shelter, siltation, engineering

RULE -> tendency, not universal binary

ASCII MASTER FLOW – PANEL 9/12: NCCR evidence boundary

PERIOD -> 1990-2018 assessment

ERODING -> about 33.6 percent

ACCRETING -> about 26.9 percent

STABLE -> about 39.6 percent; not an annual rate

ASCII MASTER FLOW – PANEL 10/12: CRZ implementation ladder

NOTIFICATION -> category-specific national rules

MAPPING -> HTL/LTL and technical inputs

CZMP -> consultation + approval + local map

PROJECT -> appraisal, decision, compliance

CLOSE DISTINCTION: Emergent and submergent coasts describe relative sea-level histories, erosional and depositional forms can coexist, and CRZ categories/regulatory lines are legal-planning constructs rather than geomorphic shoreline types.

ASCII MASTER FLOW – PANEL 11/12: Coastal response hierarchy

AVOID -> keep new exposure out of high-risk space

RESTORE / ACCOMMODATE -> buffers + resilient use

RETREAT -> move assets where persistence fails

PROTECT -> selective works with downdrift audit

EVIDENCE LIMIT: Erosion control can shift risk downdrift; ports, dams, sand mining, storms, subsidence and sea-level rise interact, while CRZ rules, clearances and project status require exact notification/date/source discipline.

ASCII MASTER FLOW – PANEL 12/12: Coast answer spine

DEFINE + MAP -> process and coastal sector

DERIVE -> energy, sediment and sea-level chain

VERIFY -> period, category, plan and status

VERDICT -> integrated, adaptive and locally mapped